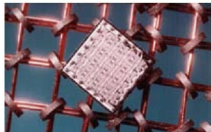


vibration monitoring. Low-voltage operation and low power consumption make it ideal for battery-powered products. It has a power-saving shut-down mode and wakes up automatically when motion is detected.

Radio chip mimics human ear

MIT engineers have built a fast, ultra-broadband, low-power radio chip, modelled on the human inner ear, that could enable wireless devices capable of receiving cell phone, Internet, radio and television signals. Rahul Sarpeshkar, associate professor of electrical engineering and computer science, and his graduate student, Soumyajit Mandal, designed the chip to mimic the inner ear, or cochlea. The chip is claimed to be faster than any human-designed radio-frequency spectrum analyser and also operates at



MIT radio chip

much lower power.

The engineers have filed for a patent to incorporate the RF cochlea in a universal or software radio architecture that is designed to efficiently process a broad spectrum of signals including cellular phone, wireless Internet, FM and other signals.

The RF cochlea mimics the structure and function of the biological cochlea, which uses fluid mechanics, piezoelectrics and neural signal processing to convert sound waves into electrical signals that are sent to the brain.

As sound waves enter the cochlea, these create mechanical waves in the cochlear membrane and the fluid of the inner ear, activating hair cells (cells that cause electrical signals to be sent to the brain). The cochlea can perceive a 100-fold range of frequencies—in humans,



Freescale's smartbook

from 100 to 10,000 Hz. Sarpeshkar used the same design principles in the RF cochlea to create a device that can perceive signals at million-fold higher frequencies, which includes radio signals for most commercial wireless applications.

Future smartbook prototypes

Freescale recently displayed next-generation 'smartbook' form factors at *Computex* show in Taiwan. The new class of devices sport larger screens than traditional smartphone devices, and are cloud-computing-centric, with all-day battery life and instant-on functionality. These are powered by processors based on ARM technology, such as Freescale's i.MX51 device.

Freescale engaged with Savannah College of Art and Design (SCAD) to explore requirements related to ergonomic issues, user interfaces, alternative form factors and even accessories for next-generation smartbook devices. This gave Freescale valuable insight into how end-users prefer to interact with smartbooks. The knowledge gained will be fed back into its chip design processes, ultimately resulting in future i.MX processors that enable compelling consumer experiences and entirely new classes of consumer devices.

New Android-powered smartphones

Huawei has come up with two Android-powered smartphones. The new

phones—U8230 and C8000—utilise the Windows Mobile operating system and will be commercially released in the second half of 2009.

The U8230 allows users to switch between full and half QWERTY keyboards and to navigate programmes easily via an 8.9cm (3.5-inch) LCD touch screen. It uses the Google Chrome Internet browser which supports a variety of Google applications such as Google Map, Google Search and Google Talk. It is also fitted with a battery which has a capacity of 1500 mAh—the longest battery capacity among Android-powered handsets currently available.

The C8000, which supports Windows Mobile operating software, incorporates a 3D menu graphic user interface and an independent display adaptor which enables the smooth presentation of video files and DVD-quality pictures on a 7.1cm (2.8-inch) touch screen. With optimised power management, it is capable of playing video files continuously for six to eight hours.

Smaller, lighter PlayStation

Watch out for a smaller and lighter PlayStation Portable (PSP) handheld game machine later this year. Sony has dubbed the new device as PSP Go. It is 43 per cent lighter than the current model but comes with a 9.7cm (3.8-inch) display, compared with a 10.9cm



Sony PlayStation Portable